ECE 410 Lab 2: Finite State Machines

Names: Dion Walton, Anahita Houshyar

CCIDs: ddwalton, ahoushya

Lab Section: D31

Abstract

In this lab, a Tausworthe pseudo-random number generator (PRNG), a 5-state Moore finite state machine (FSM) for a secure element chip, and a reverse-engineered Mealy FSM sequence detector were designed, simulated, and deployed on a Zybo-Z7 FPGA board. The initial secure element FSM was upgraded to incorporate an asynchronous reset, a 1 Hz flashing red LED alarm state, and a 15-second automatic sleep countdown timer driving a 7-segment display. Hardware operation was verified using an Analog Discovery 2 oscilloscope and on-board peripherals. Simulation testbenches achieved a 100% pass rate across all 37 test cases. Statistical analysis of 147016 output words from the Tausworthe PRNG yielded a Shannon entropy of 7.9997 bits/byte, confirming near-ideal uniform pseudo-randomness.

Design

Tausworthe Generator Design

The Tauseworthe generator consisted of 3 `taus88` submodules. Each submodule acts as a linear feedback shift register (LFSR), which is essentially a D Flip-Flop that does some combinational operation to determine the next register value. Using generics, each submodule is instantiated with a "seed" - the initial value held in the LFSR (`g_SEED`), 3 shift values (`g_SHIFT1`, `g_SHIFT2`, and `g_SHIFT3`), and a "mask" (`g_MASK`). The submodule then does the following computation (although, the code uses type casting and `shift_left()` / `shift_right()` functions instead of using the overloaded `SLL` and `SRL` operators for compliance with modern VHDL).

```
reg <= (((reg SLL g_SHIFT1) XOR reg) SRL g_SHIFT2)
        XOR
        ((reg AND g_MASK) SLL g_SHIFT3);
```

Using 3 of these submodules initialized with different seeds, shift values, and masks and XORing their registers, we get a pseudo-random number generator (PRNG) that is updated on each CLK cycle.

Secure Element FSM: Initial Design

The initial design of the secure element FSM is implemented as a Moore FSM, where the current output only depends on the current state. The state-transition diagram can be seen below in the "Truth Table & Functional Block Diagrams" section. This was done by creating a `secure_element_fsm.vhd` entity, with input ports being each of the 7 inputs, (`attack_detected`, `busy`, `self_test`, etc.) and output ports being the the RGB values that indicate the current state. The RGB values are directly mapped to an enumerated type that describe the state, where the transitions between states are defined in nested if-else blocks within a process. Simply put, a control signal `state` drives the output RGB based on the `transition` process on the rising edge of the clock. Additionally, this version of the FSM utilizes a synchronous reset, which only resets the state back to `STARTUP_STATE` on the rising edge of the clock while `rst` is held high.

The top-level module for the first iteration of the secure chip FSM (`fsm_top.vhd`) utilizes the `manual_clock` to drive the state transitions, so that testing on-board allowed more precise control.

Secure Element FSM: Upgraded Design

The upgraded design of the secure element FSM required 3 new features: an asynchronous reset, a blinking red LED on `ALARM_STATE`, and a 15-second timer that automatically transitioned the state from `SLEEP_STATE` to `IDLE_STATE` once it reached 0. To create the asynchronous reset, a second iteration of the FSM was created as `secure_element_fsm_async.vhd`. This has the same code, except the process now has `rst` in its sensitivity list (as opposed to just `clk`), while taking the `if (rst = '1')` out of the if statement for the `if rising_edge(clk) = '1'` and making the rising edge if an elsif. The change can be seen below:

```
-- asynchronous reset process structure
transition : PROCESS(clk, rst)
BEGIN
    IF rst = '1' THEN
        state <= STARTUP_STATE;
    ELSIF rising_edge(clk) THEN
        -- state transition logic
    END IF;
END PROCESS transition;
```

After adding the asynchronous reset, the remaining features were implemented by creating a higher-level entity (`fsm_wrapper.vhd`) that uses the FSM as a component. This higher-level entity takes in the control signals as input and plugs them into the ports of the FSM, while additionally having a secondary `wakeup` control signal that, if '1', acts like `request`, sending the state from `SLEEP_STATE` to `IDLE_STATE`. Additionally, the out ports of the are overridden with internal signals from the top-level entity to properly control the LED.

The 1 Hz blinking LED was implemented by taking the output of the FSM (which is known due to the `rgb_fsm_out` being a one-to-one mapping to states) and toggling the actual LED between "000" (off) and "100" (red) based on a 1 Hz clock signal only if `rgb_fsm_out = "100"` (`ALARM_STATE`). The 1 Hz clock was obtained from a `clock_divider` component which uses the system clock of 125 MHz as input. The clock divider operates by counting up on each rising edge of the system clock until it reaches `125_000_000 / 2` clock cycles, where it subsequently toggles the output clock high or low. This results in an overall 1 Hz output clock.

The counter was implemented by using the `IEEE.NUMERIC_STD` library, which allowed an 4-bit unsigned `count` timer to be subtracted by 1 each rising edge of the 1 Hz clock (or where the closest rising edge in line with the system clock is). Similarly to the red flashing led, we can reuse the same logic and only count down if the FSM is in `SLEEP_STATE`. Once the `count` signal reaches 0, the `wakeup` signal is asserted, overriding normal operation and putting the FSM into `IDLE_STATE`.

The top-level implementation uses the count signal as a port that drives a seven-segment display decoder (`display_controller`) by converting the hex value to binary encoded decimal to properly display the sleep countdown value in base-10.

Black Box Design

The black box is a reverse-engineered version of the provided `black_box.bit` bitstream file. We were able to determine that the original system had 8 total states (numbered 0-7). In each state, `SW0` (indicated on `LD0`) on the Zybo-Z7 board determines where the next state will be. This "next state" is then displayed on the LEDs `LD[1:3]`, and advanced to only when `BTN0` (`clk_en`) is pressed.

To implement this, a look-up table for each state was created. The inputs for the look-up table were the current state, as well as the current input. This determined what the next state would be. Implementing the `clk_en` button for state advancement was much more difficult than a simple button debounce on a micro-controller. Since button presses have some transients and high frequency noise for some short time, the `clk_en` signal from the button had to be sampled at a lower frequency to avoid multiple edge detections. In code, we tuned the sampling frequency to be 10 Hz by counting to 12 500 000 (incrementing by one each system clock cycle). 10 Hz was chosen arbitrarily, as humans typically press buttons for longer than 1/10 seconds. However, this didn't get rid of the problems. Since button inputs are inherently in a different clock domain than the system clock, it is possible to get metastable signals (neither 0 nor 1). Hence, to account for this, a 2-stage synchronizer was used to ensure the possibly metastable input decayed towards 0 or 1 before incrementing the counter or using the signal to detect its rising edge.

Calculations/Assumptions

Secure Element FSM Assumptions

For the secure element FSM, it was assumed that any state (aside from `SLEEP_STATE`) immediately goes into `ALARM_STATE` once `attack_detected` is asserted. Additionally, it was assumed that `ALARM_STATE > SECURE_STATE > STARTUP_STATE > SLEEP_STATE` in terms of precedence (i.e., if each signal were asserted at the same time, the FSM would go into `ALARM_STATE`). Finally, it was assumed that the `SLEEP_STATE` can only be left by asserting `request`, where it transitions into `IDLE_STATE`.

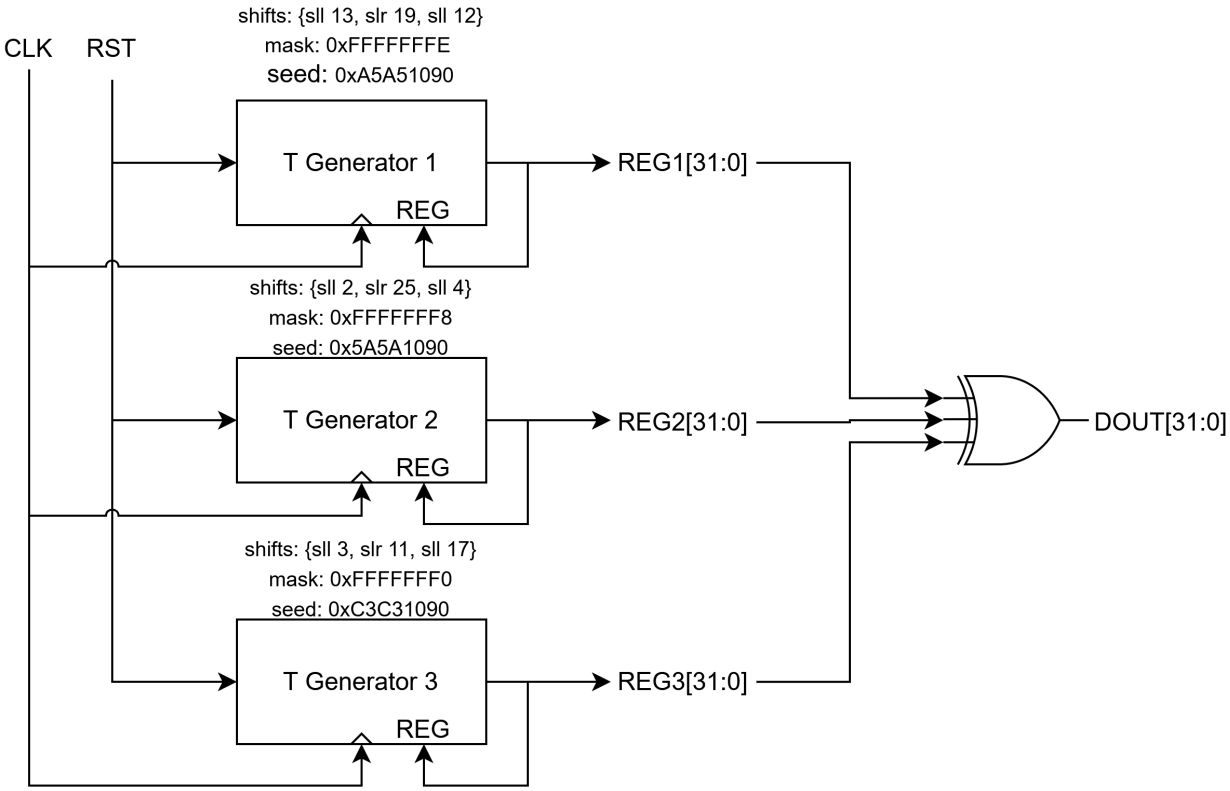
Black Box Assumptions

For the "black box," it was assumed that the original code operated as a Mealy machine (inputs and state in conjunction controlling the current state). Although mathematically analogous, the Mealy implementation was far simpler to implement than a Moore machine due to the number of states it would require.

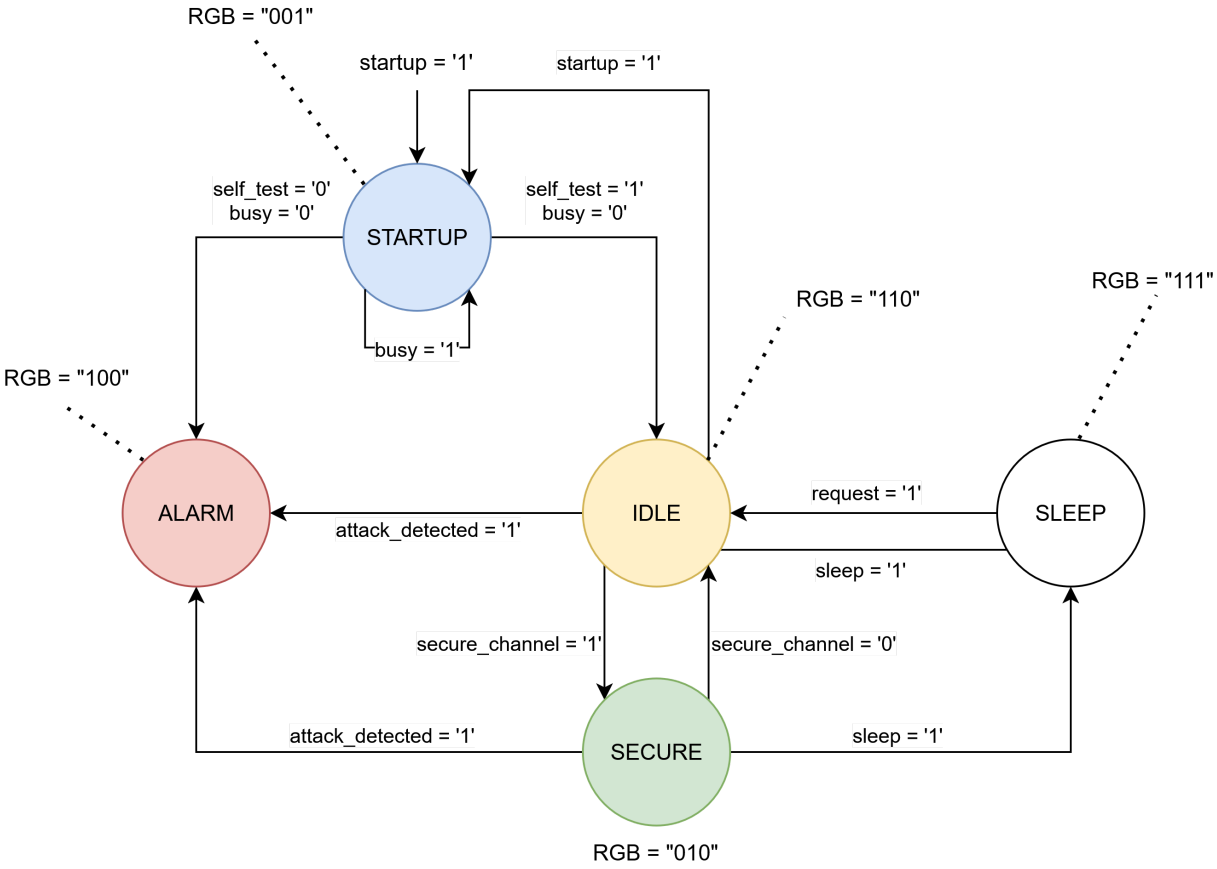
System Architecture & Block Diagrams

Truth Table & Functional Block Diagrams

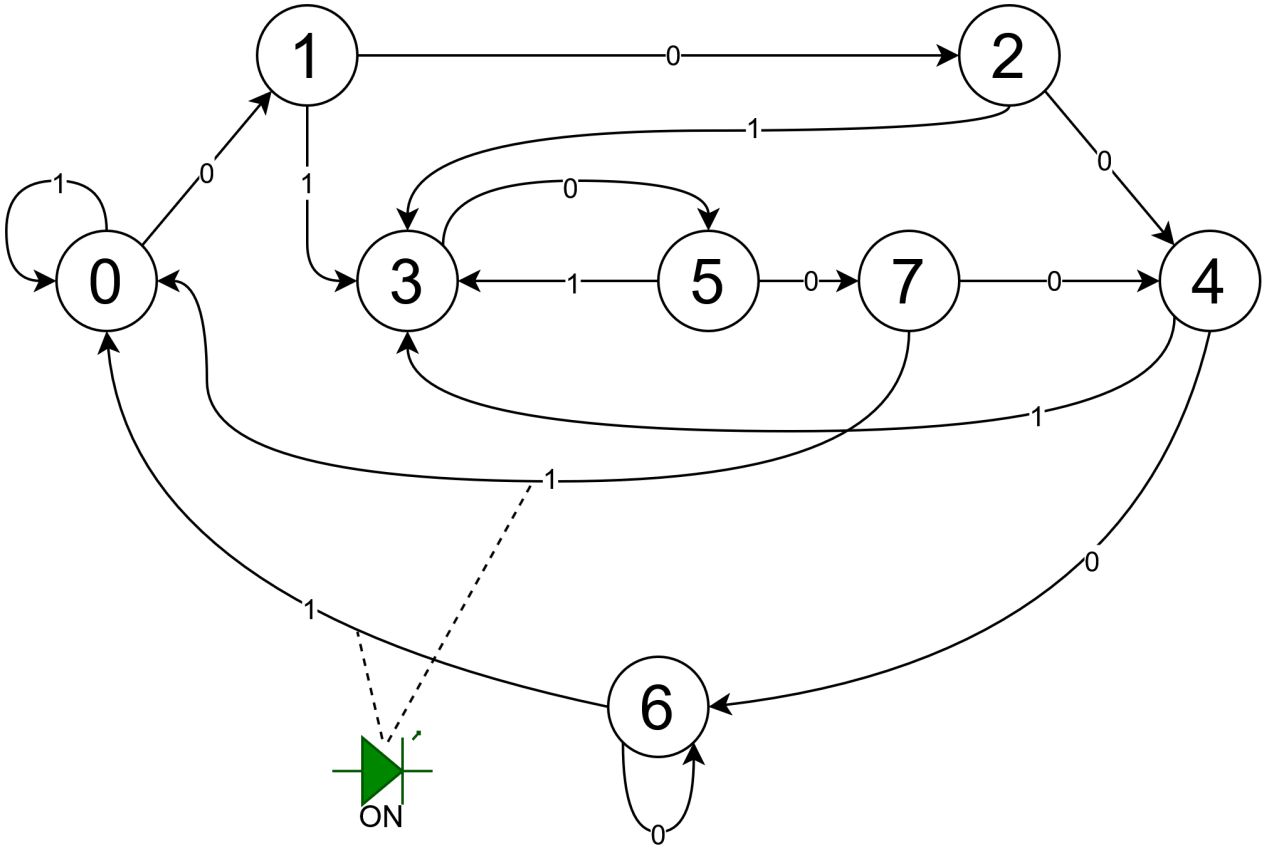
Tausworthe Generator Block Diagram



Secure Element Chip FSM



Block Box FSM



Black Box Truth Table

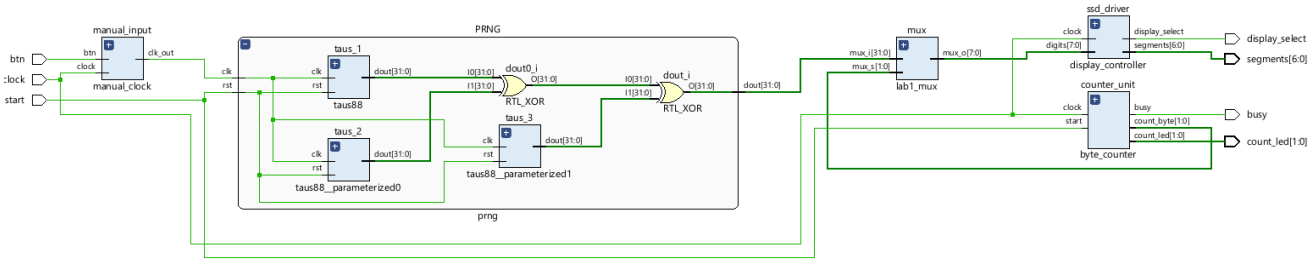
Current State (curr_state)	Input (data_in) / SW0 = LD0	Next State (next_state) / LD[1:3]	Output (seq_detected)
000	0	001	0
000	1	000	0
001	0	010	0
001	1	011	0
010	0	100	0
010	1	011	0
011	0	101	0
011	1	000	0
100	0	110	0
100	1	011	0
101	0	111	0

Current State (curr_state)	Input (data_in) / SW0 = LD0	Next State (next_state) / LD[1:3]	Output (seq_detected)
101	1	011	0
110	0	110	0
110	1	000	1
111	0	100	0
111	1	000	1

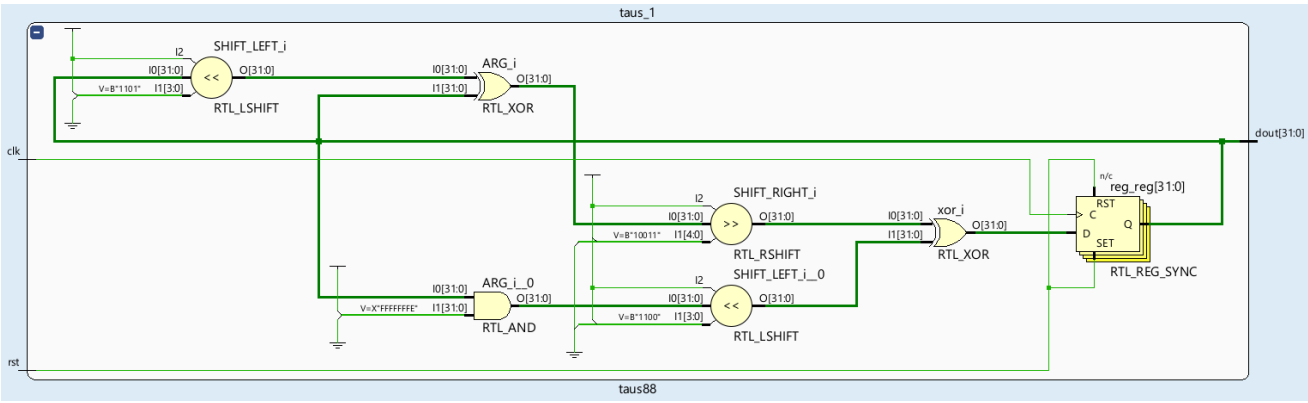
Synthesized Schematics

Tausworthe Generator RTL Synthesis

Top-Level Tausworthe Generator RTL Synthesis

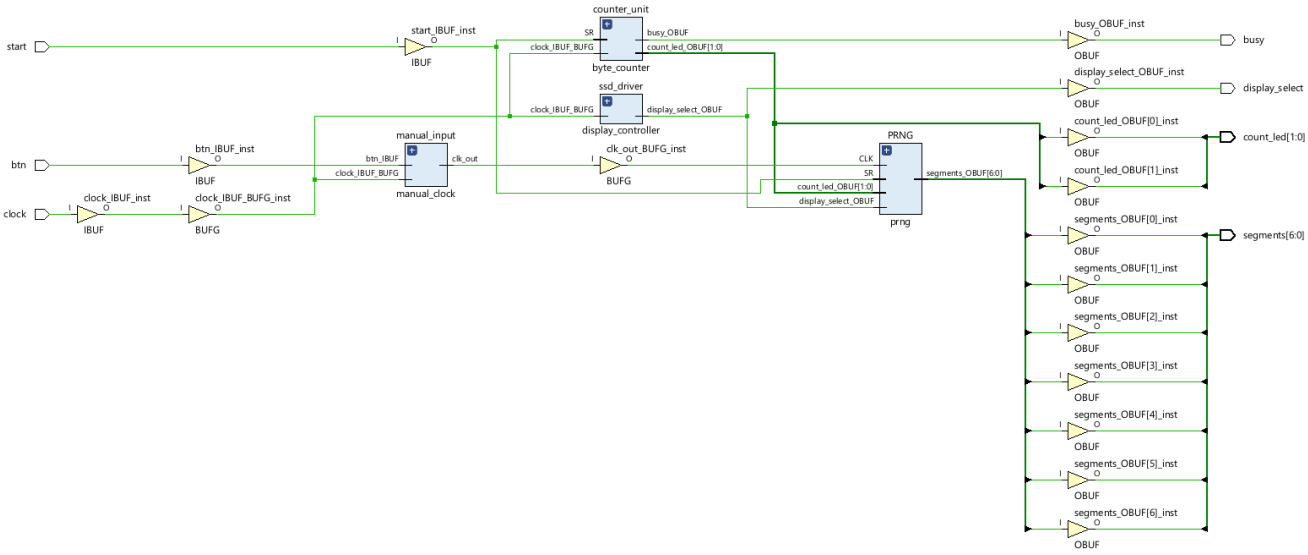


Single Instantiated taus88 Unit RTL Synthesis

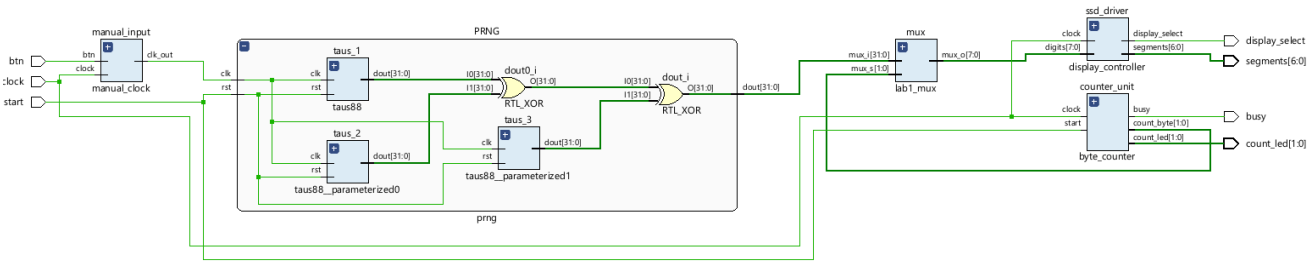


Tausworthe Generator Elaborated Design

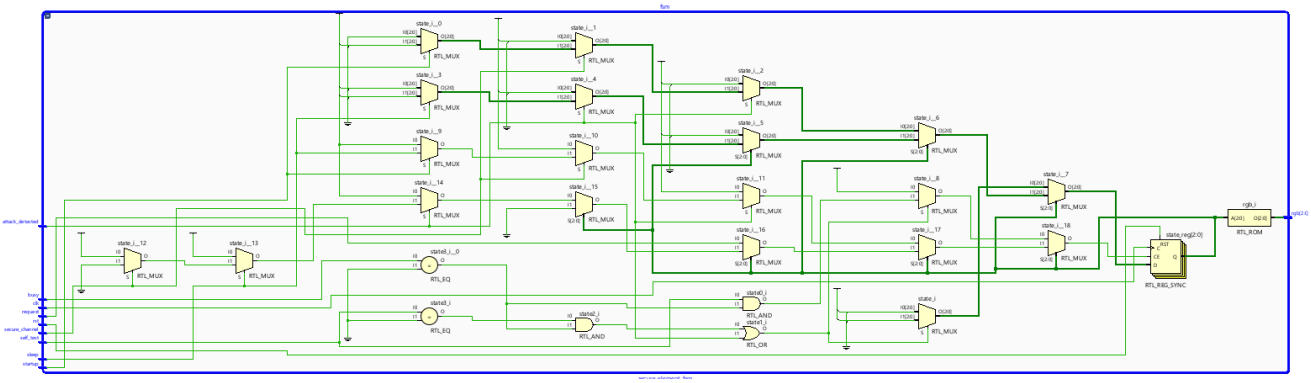
Top-Level Tausworthe Generator Elaborated Design



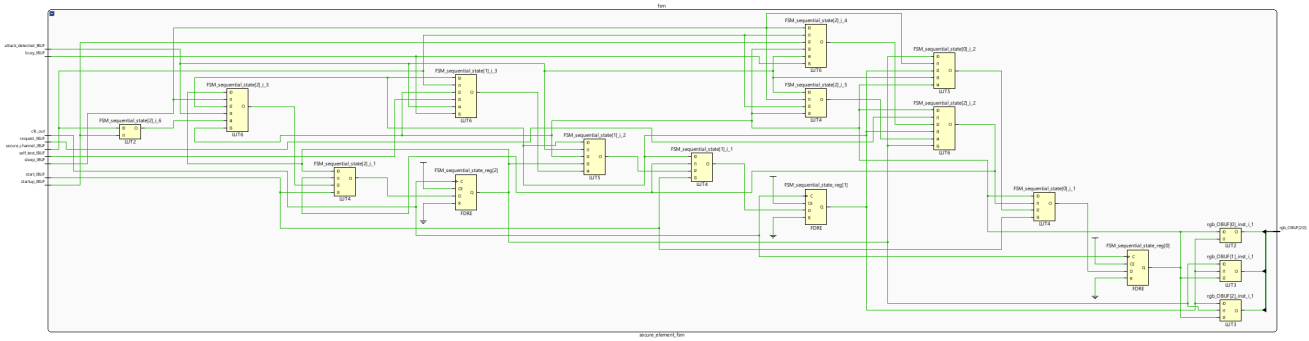
PRNG Elaborated Design



Secure Element FSM: First Version RTL Synthesis

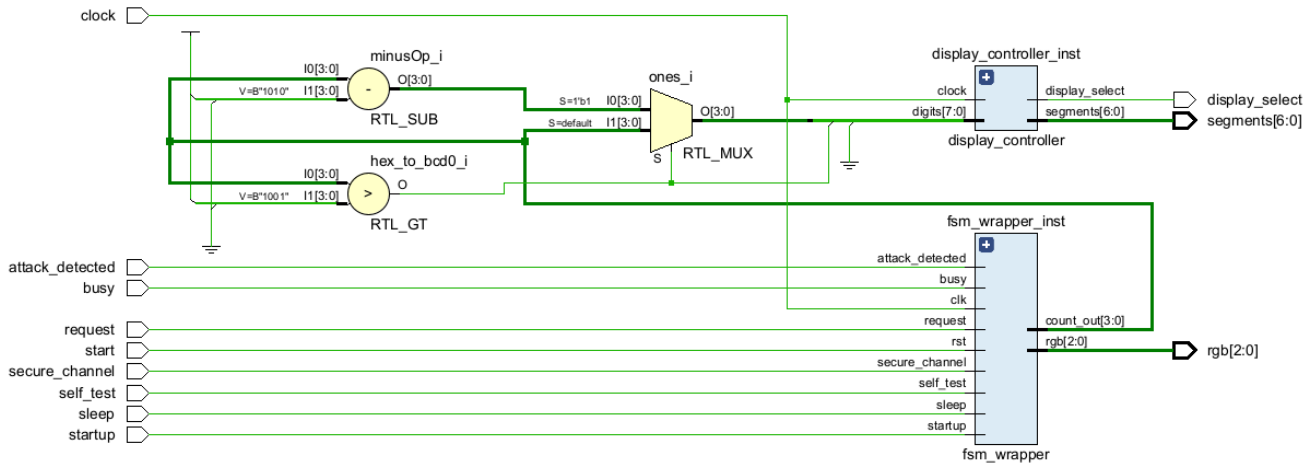


Secure Element FSM: First Version Elaborated Design

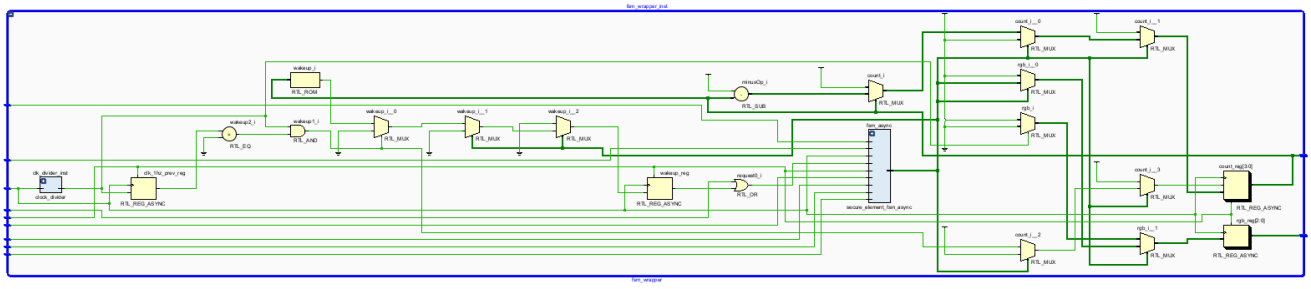


Secure Element FSM: Upgraded Version Elaborated Design

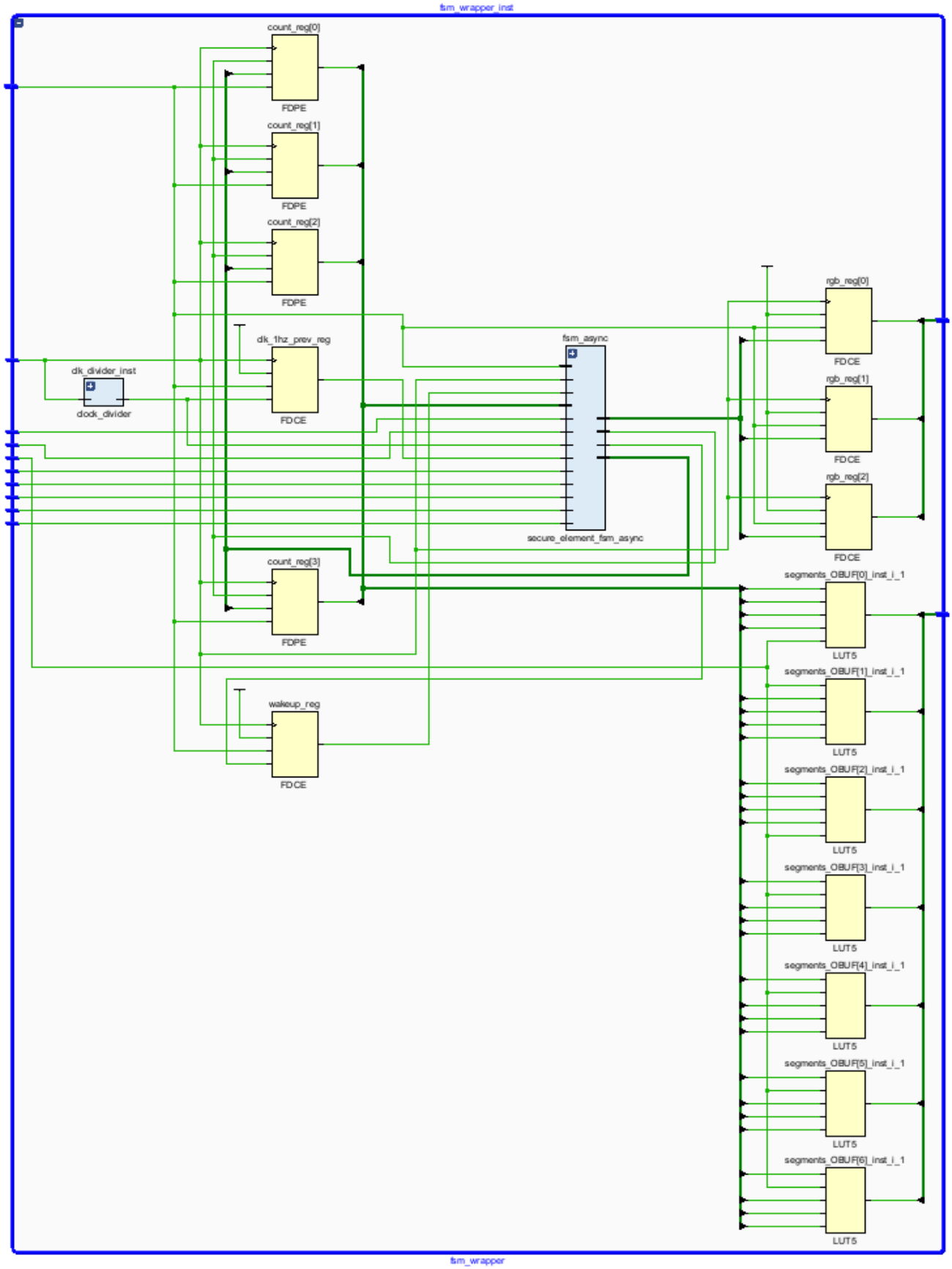
`fsm_wrapper_top` RTL Synthesis



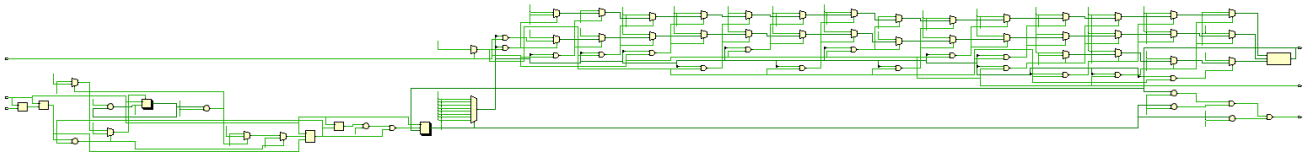
`fsm_wrapper_inst` RTL Synthesis



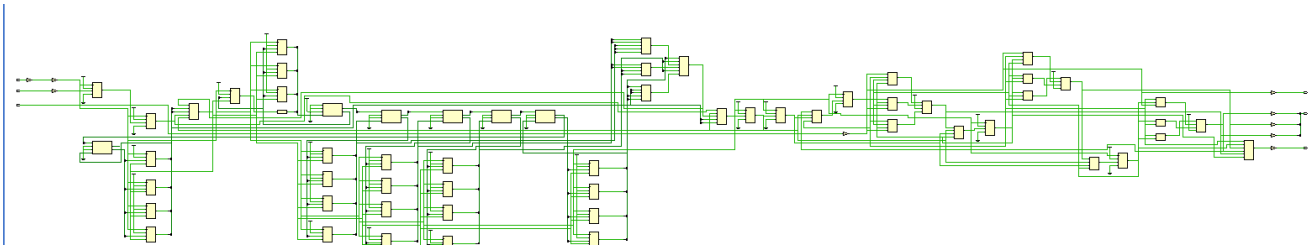
`fsm_wrapper_inst` Elaborated Design



Black Box RTL Synthesis



Black Box Elaborated Design



Implementation

Tausworthe Generator Implementation

For the Tausworthe pseudo-random number generator, the implementation utilized an instantiation of the top-level design from `prng.vhd` in `taus88_top.vhd`. This implementation utilized a button-controlled manual clock to advance cycle through each clock cycle of the PRNG, essentially generating a new number each button press. Meanwhile, a byte counter takes in the generated number and displays put it in a byte-long section, while a display controller displays the byte-long section on the seven-segment display as hexadecimal.

Secure Element FSM Implementation

The secure element FSM required taking in control signals from the Analog Discovery 2 over a Pmod Port connection. To accomplish this, the constraints file needed to be modified (seen in `fsm_constr.xdc`). A table below shows the pin assignments that were required. The first version utilized `btn` (Pin `K17`) to manually advance the state, this was removed in the second version, instead using the system clock (`clock`). Both versions use a `start` signal from the button at pin `K18` as a reset signal. Both versions also use control RGB LED using the signal `rgb[2:0]`. Lastly, the improved version uses `segments[7:0]` that map to the pins that control the seven-segment display to show the current count in `SLEEP_STATE`.

FPGA Pin	Signal Name	AD2 Signal	Description
K17	`clock`	—	System Clock (125 MHz)
K18	`start`	—	Start/Reset Button
Y16	`btn`	—	Advance State Button (For first version)

FPGA Pin	Signal Name	AD2 Signal	Description
V16	`rgb[2]`	—	RGB LED 6 (Red)
F17	`rgb[1]`	—	RGB LED 6 (Green)
M17	`rgb[0]`	—	RGB LED 6 (Blue)
V15	`busy`	**D0**	Pmod JC Pin 1
W15	`self_test`	**D1**	Pmod JC Pin 2
T11	`startup`	**D2**	Pmod JC Pin 3
T10	`sleep`	**D3**	Pmod JC Pin 4
W14	`request`	**D4**	Pmod JC Pin 7
Y14	`secure_channel`	**D5**	Pmod JC Pin 8
T12	`attack_detected`	**D6**	Pmod JC Pin 9
T14	`segments[6]`	—	Pmod JD Pin 1 (7-Segment Bit 6)
T15	`segments[5]`	—	Pmod JD Pin 2 (7-Segment Bit 5)
P14	`segments[4]`	—	Pmod JD Pin 3 (7-Segment Bit 4)
R14	`segments[3]`	—	Pmod JD Pin 4 (7-Segment Bit 3)
V12	`segments[2]`	—	Pmod JE Pin 1 (7-Segment Bit 2)
W16	`segments[1]`	—	Pmod JE Pin 2 (7-Segment Bit 1)
J15	`segments[0]`	—	Pmod JE Pin 3 (7-Segment Bit 0)
H15	`display_select`	—	Pmod JE Pin 4 (Digit / Display Select)

Black Box Implementation

TO reimplement the black box, only the system clock, input from the switch, a `clk_en` button to advance the state, 4 single-color LEDs (`led[0:3]`), and a pin for the green RGB LED was required for `seq_detected`. A table can be seen below. The AD2 was not used for the black box.

FPGA Pin	Signal Name	Description
K17	`clock`	System Clock (125 MHz)
G15	`data_in`	Switch 0 (`sw[0]`)
K18	`clk_en`	Button 0 (`btn[0]`)
M14	`data`	LED 0 (`led[0]`)

FPGA Pin	Signal Name	Description
M15	`leds[0]`	LED 1 (`led[1]`)
G14	`leds[1]`	LED 2 (`led[2]`)
D18	`leds[2]`	LED 3 (`led[3]`)
F17	`seq_detected`	RGB LED 6 Green (`led6_g`)

Simulation

Test Cases Table & Explanation

Tausworthe Generator Test Cases

The Tausworthe PRNG did not require any complex test cases, instead its functionality was verified by generating a clock to cycle through each value of the LFSR and testing that the reset signal set the register back to initial value. Since each

Test Case	Expected Output	Actual Output	Pass / Fail	Description
Initialization	`0x3C3C1090`	`0x3C3C1090`	Pass	Verified initial state corresponds to XOR of seeds
Next Clock Edge	`0xD593B29D`	`0xD593B29D`	Pass	Verified output updates correctly on rising clock edge
Reset Asserted	`0x3C3C1090`	`0x3C3C1090`	Pass	Verified synchronous reset restores initial seed value

Secure Element FSM: First Version Test Cases

Test Case	Expected Output	Actual Output	Pass / Fail	Description
Test 1	`rgb = "001"`	`rgb = "001"`	Pass	Transition `rst` -> `startup`: System enters startup state
Test 2	`rgb = "100"`	`rgb = "100"`	Pass	Transition `startup` -> `alarm`: Failed self-test triggers alarm state
Test 3	`rgb = "001"`	`rgb = "001"`	Pass	Transition `alarm` -> `startup`: Reset clears alarm to startup state

Test Case	Expected Output	Actual Output	Pass / Fail	Description
Test 4	`rgb = "110"`	`rgb = "110"`	Pass	Transition `startup` -> `idle`: Successful self-test transitions to idle state
Test 5	`rgb = "111"`	`rgb = "111"`	Pass	Transition `idle` -> `sleep`: Sleep signal transitions FSM to sleep state
Test 6	`rgb = "110"`	`rgb = "110"`	Pass	Transition `sleep` -> `idle`: Request signal wakes FSM to idle state
Test 7	`rgb = "001"`	`rgb = "001"`	Pass	Transition `idle` -> `startup`: Startup signal returns FSM to startup state
Test 8	`rgb = "110"`	`rgb = "110"`	Pass	Transition `startup` -> `idle`: Successful self-test returns to idle state
Test 9	`rgb = "100"`	`rgb = "100"`	Pass	Transition `idle` -> `alarm`: Attack detection triggers alarm state
Test 10	`rgb = "001"`	`rgb = "001"`	Pass	Transition `alarm` -> `startup`: Reset signal clears alarm state to startup
Test 11	`rgb = "110"`	`rgb = "110"`	Pass	Transition `startup` -> `idle`: Successful self-test enters idle state
Test 12	`rgb = "010"`	`rgb = "010"`	Pass	Transition `idle` -> `secure`: Secure channel request enters secure state
Test 13	`rgb = "111"`	`rgb = "111"`	Pass	Transition `secure` -> `sleep`: Sleep signal from secure state enters sleep state
Test 14	`rgb = "110"`	`rgb = "110"`	Pass	Transition `sleep` -> `idle`: Request signal wakes FSM back to idle state
Test 15	`rgb = "010"`	`rgb = "010"`	Pass	Transition `idle` -> `secure`: Secure channel asserted enters secure state
Test 16	`rgb = "110"`	`rgb = "110"`	Pass	Transition `secure` -> `idle`: De-asserting secure channel returns to idle state
Test 17	`rgb = "010"`	`rgb = "010"`	Pass	Transition `idle` -> `secure`: Re-asserting secure channel enters secure state
Test 18	`rgb = "100"`	`rgb = "100"`	Pass	Transition `secure` -> `alarm`: Attack detected in secure state triggers alarm

Secure Element FSM: Upgraded Version Test Cases

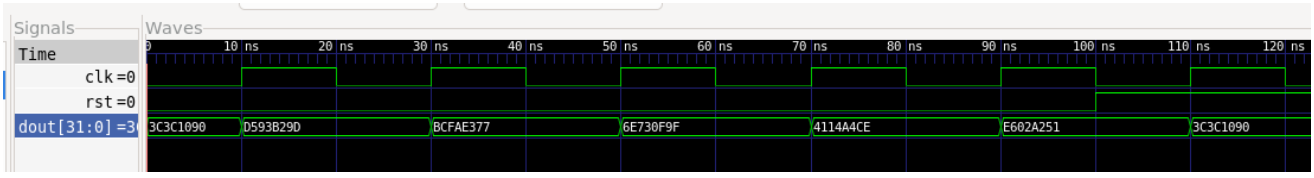
Test Case	Expected Output	Actual Output	Pass / Fail	Description
Test 1	`rgb = "001"`	`rgb = "001"`	Pass	Transition `rst` -> `startup`: System enters startup state
Test 2	`rgb = "100"` or `"000"`	`rgb = "100"`	Pass	Transition `startup` -> `alarm`: Alarm state entered with red LED output
Test 3	`count_out = "1111"`	`count_out = "1111"`	Pass	Alarm state verification: `count_out` held reset at 15
Test 4	`rgb = "100"` or `"000"`	`rgb = "000"`	Pass	Alarm state verification: Red LED flashes with 1 Hz clock divider
Test 5	`rgb = "001"`	`rgb = "001"`	Pass	Transition `alarm` -> `startup`: Reset clears alarm state to startup
Test 6	`rgb = "110"`	`rgb = "110"`	Pass	Transition `startup` -> `idle`: Successful self-test transitions to idle state
Test 7	`rgb = "111"`	`rgb = "111"`	Pass	Transition `idle` -> `sleep`: Sleep signal enters sleep state with white LED
Test 8	`rgb = "111"`	`rgb = "111"`	Pass	Sleep state verification: RGB remains solid white during countdown
Test 9	`count_out < "1111"`	`count_out = "1110"`	Pass	Sleep state verification: Counter decrements on clock divider edges
Test 10	`rgb = "110"`	`rgb = "110"`	Pass	Transition `sleep` -> `idle`: Auto-wakeup triggers return to idle state when counter reaches 0

Black Box Test Cases

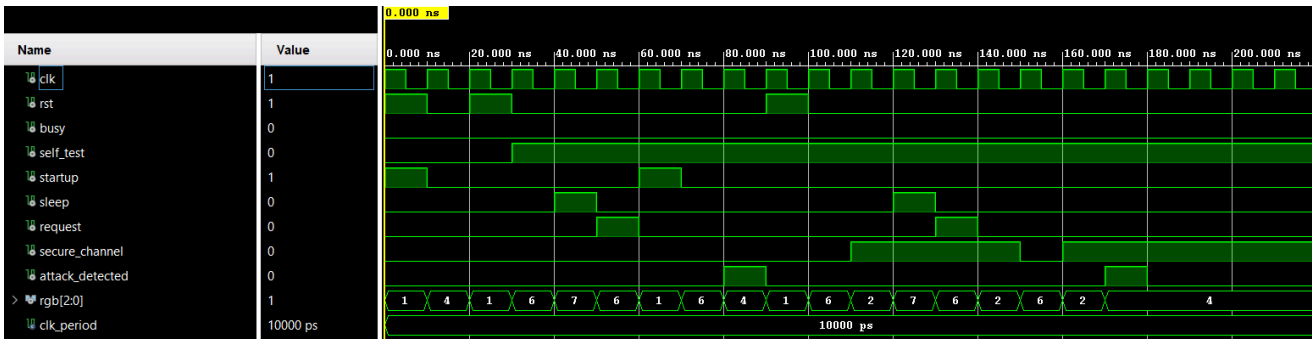
Test Case	Expected Output	Actual Output	Pass / Fail	Description
Test 1	`data = '0', leds = "001"`	`data = '0', leds = "001"`	Pass	Initial state: verified data pass-through and `next_state = 001`
Test 2	`leds = "001"`	`leds = "001"`	Pass	Glitch rejection: short pulse on `clk_en` ignored by debounce counter

Test Case	Expected Output	Actual Output	Pass / Fail	Description
Test 3	`leds = "010"`	`leds = "010"`	Pass	Transition state `000` -> `001`: debounced `clk_en` advances state
Test 4	`leds = "100"`	`leds = "100"`	Pass	Transition state `001` -> `010`: debounced `clk_en` advances state
Test 5	`leds = "110"`	`leds = "110"`	Pass	Transition state `010` -> `100`: debounced `clk_en` advances state
Test 6	`leds = "110"`	`leds = "110"`	Pass	Transition state `100` -> `110`: debounced `clk_en` advances state
Test 7	`data = '1', leds = "000", seq_detected = '1'`	`data = '1', leds = "000", seq_detected = '1'`	Pass	Sequence detection: combinatorial check in state `110` with `data_in = '1'`
Test 8	`seq_detected = '0'`	`seq_detected = '0'`	Pass	Transition state `110` -> `000`: clock pulse clears sequence detection flag
Test 9	`leds = "000"`	`leds = "000"`	Pass	Hold state `000`: `data_in = '1'` maintains current state at `000`

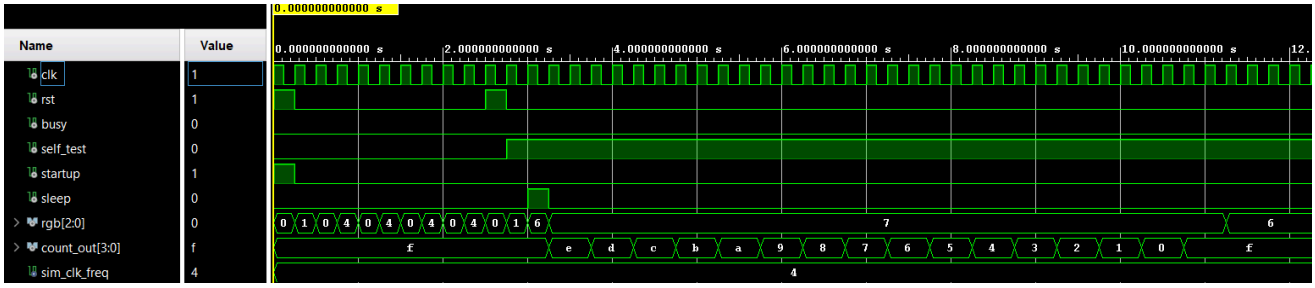
Simulation Results



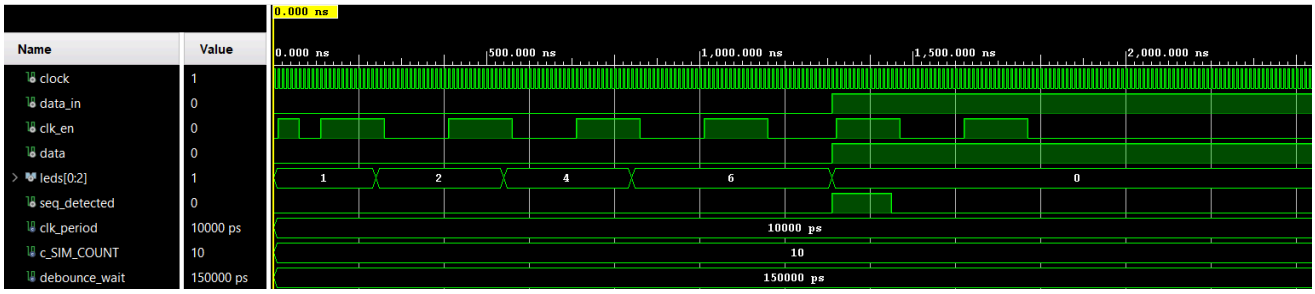
[sim. 1] The above figure are the simulation results of the Tausworthe PRNG. As can be seen during initialization ($t = 0$), the last 4 hex digits contain my student ID (176**1090**), which is expected as XORing 3 registers with `0x1090` as their LSBs results in `0x1090` in the output's LSB. It can also be seen at the end that the reset signal `rst` makes the output `dout[31:0]` the same as it's initial value.



[sim. 2] The above figure shows the simulation results of the secure element FSM through each state transition. It should be noted that this version has a synchronous reset. Each state properly outputs the `rgb[2:0]` value associated with their state.



[sim. 3] The above figure shows the simulation results of the upgraded secure element FSM. More specifically, it tests the `fsm_wrapper` entity. Thus, all that needed to be tested is that for an arbitrary clock frequency (in this case 4 Hz), the timer counted down in sleep mode (`rgb = 111`) at the desired 1 Hz (seen in `count_led` going from `0xF` = $(15)_{10}$ down to 0x0 = 0_{10}), and flashing red (`rgb` oscillating between `0b000` = 0 and `0b100` = 4).



[sim. 4] The above figure shows the simulation results for the black box re-implementation. More specifically, it focuses on the debouncing filter logic and state transition dynamics driven by the `clk_en` input. The waveform verifies that short glitches on `clk_en` lasting fewer than `g_COUNT_MAX` clock cycles are ignored, whereas a sustained high signal allows the internal counter to expire and assert `slow_clk_en`. This rising edge successfully triggers state transitions (updating `leds` via `curr_state`), and upon reaching state `"110"` with `data_in = '1'`, `seq_detected` correctly asserts high before clearing once the state resets back to `"000"`.

Results

Tausworthe PRNG: The PRNG simulation was checked based on three things: the initial output, its update after clock edge, and how it response to reset. In the initialization stage the three Tausworthe registers load their seed values and their XOR produced the initial 32-bit output, 0x3C3C1090. After the rst changes from 1 to 0 the output changes and updates on successive rising clock edges. When reset is asserted, the generator returns to those starting values and the sequence begins again.

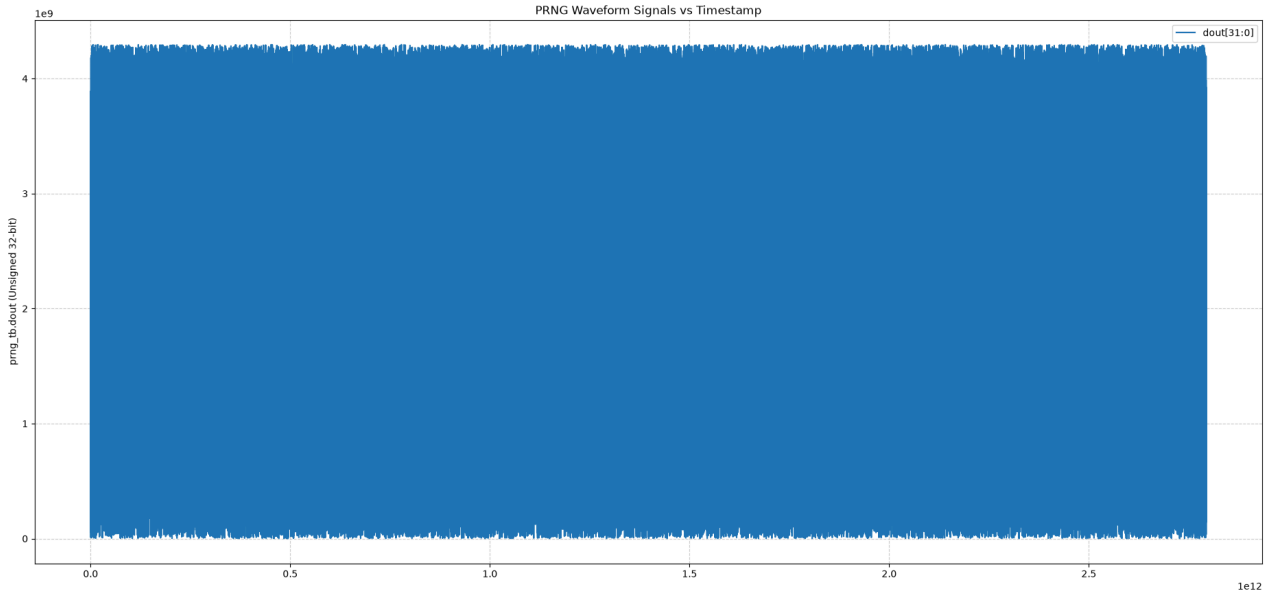
Test case	Expected result	Observed result
Initial state	`0x3C3C1090`	`0x3C3C1090`
First update after a rising clock edge	`0xD593B29D`	`0xD593B29D`
Reset asserted at a rising clock edge	Output returns to `0x3C3C1090`	Output returned to `0x3C3C1090`

All three test cases matched their expected results.

As an additional metric of PRNG success, an entropy analysis was conducted over a set of simulation data obtained by omitting the reset assertion in the testbench. 147016 32-bit values were created. Using a vcd2csv converter (available: GitHub) and a python program to test the Shannon entropy (see equation below) of each individual bit, the calculated byte entropy was found to be 7.9997 bits/byte (8 bits/byte would be ideal) -- which essentially means it would require just barely under 8 bits to compress the entire 147016 word long PRNG sequence. These results are quite good, and show that there is little to no biasing towards 0 or 1, or repeating patterns that can be extracted. This is indicative of a good, uniformly distributed set of values.

Shannon entropy formula:

$$H(X) = -\sum_{x \in X} p(x) \log(p(x))$$



The PRNG generated value (encoded as a 32-bit unsigned integer) vs. time can be seen in the graph above.

Secure Element FSM Results: This section was simulated to verify the FSM transitions between the startup, idle, secure, sleep, and alarm states. The RGB outputs identify the current state. `001` for startup, `110` for idle, `010` for secure, `111` for sleep, and `100` for alarm. In the simulation, the observed RGB output matched the expected output in all 18 test cases.

State	RGB output	Indicated state
Startup	`001`	Blue
Idle	`110`	Yellow
Secure	`010`	Green
Sleep	`111`	White
Alarm	`100`	Red

All 18 test cases passed, with no differences between the expected and observed outputs.

Upgraded Secure Element FSM Results: The upgraded FSM was evaluated for its asynchronous reset, alarm LED flashing, and sleep countdown.

Feature	Expected behavior	Observed behavior
Asynchronous reset	Return to startup when reset is asserted, without waiting for a clock edge	The FSM returns to startup when reset is asserted
Alarm indicator	Flash the red LED at 1 Hz while in the alarm state	The LED alternates between red (`100`) and off (`000`)

Feature	Expected behavior	Observed behavior
Sleep countdown	Count down from 15 and wake the FSM to idle	the wakeup signal returns the FSM to idle

Sequence Detector Results: It was simulated to check its state transitions, data pass-through, sequence detection, and rejection of short clk_en pulses. The design has eight states, and the LEDs show the next state. No discrepancies were observed in these tests. The initial test confirmed that `data_in = '0'` passed through to `data` and that the LEDs displayed `001`. The detector then advanced through the tested states: `000` to `001`, `001` to `010`, `010` to `100`, and `100` to `110`, with the LEDs showing the expected outputs at each step.

For sequence detection, when the detector was in state `110` and `data_in = '1'`, the output became `data = '1'`, the LEDs displayed `000`, and `seq_detected` was changed to 1. After the detector moved to state `000`, `seq_detected` returned to `0`.The observed results matched the expected results in all nine tests.

Conclusion

In this lab we worked on designing and testing a Tausworthe pseudo-random number generator, a secure element FSM, upgrades to the FSM, and a reverse-engineered sequence detector. The simulation results matched the expected outputs for all three PRNG tests, all 18 secure element FSM tests, and all sequence detector tests. The hardest part was making sure each `clk_en` button press advanced the sequence detector only once. The FSM requirements also needed clarification through repeated design updates. For the PRNG, the SRL and SLL operators did not work as expected with STD_LOGIC_VECTOR, so we used shift_right and shift_left with unsigned type conversions. The setup could be improved by using secure cables to connect the Zybo board and breadboard to the Analog Discovery 2 oscilloscope, since loose connections made the setup less reliable.

Questions

> 1. What was the biggest obstacle you had to overcome while working on this lab, and how did you resolve it?

As stated earlier, the biggest obstacle in this lab was certainly the `clk_en` button debouncing. Here, we had to deal with asynchronous clock domains (the real world vs. the system clock). In software, its quite easy: just trigger an interrupt when a button is pushed to wake a task or simply poll the button at a slower rate using a `delay_ms(10)`. To deal with metastable signals in hardware, it requires slow sampling and double flopping to avoid glitches, which usually is abstracted away from a programmer.

> 2. Compare the testing process using the logic analyzer with the behavioral simulation. How did the real-time testing with the AD2 differ from the purely simulated environment, and what are the key benefits of using the AD2/AD3 for

> testing your design? Were there any specific challenges or insights you gained from this hands-on testing that were not apparent during simulation?

Real-time testing with the AD2 was quite similar to the behavioural test in simulation, except it allowed for quick and easy changes to the system inputs manually via the WaveForms GUI rather than having to hand-code each possible transition by hand. One key observation made is that keeping some logic inputs high by accident may lead to surprising, although intended consequences (such as triggering `SLEEP_STATE` almost immediately after switching into `IDLE_STATE` if `sleep` is held high). Using the AD2 honestly gave no further insights than the behavioural simulations or testing using other pins could have, aside from providing occasionally faulty signals. If I were to improve this lab, I would scrap the AD2 for lab 2 and perhaps use one of the keypads with an entity to push/pull signals to avoid the risks and wire spaghetti that comes with the AD2.

> 3. Propose potential enhancements to the pseudo random number generator design from Part 1, and briefly justify each enhancement in 1 or 2 sentences.

The PRNG can be improved by adding another component to the LFSR and upgrading to the taus113 architecture. This increases the total period to $2^{113} - 1$ cycles from the taus88's $2^{88} - 1$.

> 4. Suggest other possible upgrades that could be added to the secure element to improve its functionality or safety. Provide a brief explanation for each suggestion.

The secure element could be improved by making a new state that triggers after some amount of consecutive `attack_detected` signals without going into `SECURE_STATE`. This might disable the `rst` entirely and require some other process to recover. Additionally, it could be improved by using a handshaking protocol instead of an open-loop signal that triggers `SECURE_STATE`.

> 5. What trade-offs did you consider when deciding whether to use a Mealy or Moore FSM for the secure element? How did this decision impact your design?

A Mealy machine was an obvious choice fpr the secure element FSM as it no outputs depended on current inputs, rather, the outputs (`rgb` values) were determined by purely the state. This made the design simple and states mutually exclusive from eachother while also avoiding multiple LUTs depending on state and input. The black box could have been implemented as a Moore machine, but the amount of states it would require would be enormous compared to the actual design we have. However, it did require a tedious amount of tests for the 8 states that were devised, as each state had to be tested for `data` = 1 and `data` = 0.

References

[1] M. Mézard and A. Montanari, *Information, Physics, and Computation*, Part A. Draft manuscript. [Online]. Available:

https://web.stanford.edu/~montanar/RESEARCH/BOOK/partA.pdf

[2] P. L'Ecuyer, "Maximally equidistributed combined Tausworthe generators," *Math. Comput.*, vol. 65, no. 213, pp. 203–213, 1996.bit vectors with a period length of $2^{88} - 1$.

[3] P. L'Ecuyer, "Tables of maximally equidistributed combined LFSR generators," *Math. Comput.*, vol. 68, no. 225, pp. 261–269, 1999.

Team Contribution

Peer programming and debugging was used in this lab, so exact numbers or percentages are not provided. Report was a collaborative process.

Dion Walton: Design, simulation, implementation, report writing.

Anahita Houshyar: Design, simulation, implementation, report writing.